R16

Code No: 138GK

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2023
MICROPROCESSORS AND MICROCONTROLLERS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What will happen if a label within a macro is not declared local?	[2]
b)	Discuss about the 8086 interrupt types.	[3]
c)	What are the Addressing Modes for a Micro Controller?	[2]
d)	How many ports are available in 8051 micro controller? What are they?	[3]
e)	What is SPI bus?	[2]
f)	Describe about LCD display of 8051 microcontroller.	[3]
g)	Differentiate ARM and Thumb instruction set features.	[2]
h)	Give different applications of ARM processors.	[3]
i)	List the registers used in OMAP Processor.	[2]
j)	Briefly discuss the pin diagram of CORTEX processor.	[3]

PART – B

(50 Marks)

2.a) Explain the register organization of 8086.
b) Explain the physical memory organization of 8086 system. [5+5]

OR

3.a) Discuss in detail the various signals of 8086.
b) Write an 8086 assembly language program to sort an array of data in descending order. [5+5]

4.a) Explain in detail about the special function register of 8051 in detail.
b) Explain the instruction set of 8051. [5+5]

OR

5.a) Write an ALP to multiply the given number 48H and 30H using 8051 microcontroller.
b) Explain about the Timers of 8051 and the Registers used for 8051 Timers. [5+5]

6.a) With neat sketch, explain the function of external memory RAM and ROM with 8051 microcontroller?
b) Discuss in detail about RS232 interface. [5+5]

OR

7.a) With neat sketch, explain the function of DAC with 8051 microcontroller?
b) Explain in detail about USB communication interface. [5+5]

8.a) With a neat diagram, explain the different general purpose registers of ARM processors.
b) Mention about the program status register instructions in ARM processor. [5+5]

OR

9.a) Write ARM data processing Instructions.
b) Explain PSR Instructions with examples. [5+5]

10.a) Briefly discuss about the interrupts in CORTEX processor.
b) Design the block diagram for OMAP processor. [5+5]

OR

11.a) Explain Memory processing and commands used in OMAP processor.
b) Write a Simple Program using Instructions of CORTEX processor. [5+5]

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